# FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Biology — Class IX | Paper: SSC-I

**Syllabus: Ch 1 — The Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle | Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec 7.1–7.6)**

**Total Marks: 65** **Time: 2h 30m** **Version: 4**

**General Instructions:**

1. Write your Roll Number in the space provided on the answer sheet. Do NOT write your name.
2. Attempt ALL questions from Section A. In Sections B and C, attempt the required number choosing either the main question OR its alternative (OR).
3. All parts of a question must be answered in sequence and at one place.
4. Extra attempt of any question or any part thereof will NOT be considered.

## SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

| # | Question | A | B | C | D | Answer |
|---|---|---|---|---|---|---|
| 1 | Dengue fever (unlike malaria, spread by *Anopheles*) is spread by a mosquito with zebra-like black and white body stripes. This mosquito is: | Anopheles | Culex | Aedes aegypti | Plasmodium | ⒶⒷⒸⒹ |
| 2 | A scientist who studies effects of low temperature on living organisms, to safely preserve cells and tissues, is a(n): | Bioinformatician | Biomedical engineer | Astrobiologist | Cryobiologist | ⒶⒷⒸⒹ |
| 3 | Which early biologist (1656–1708), before Linnaeus, introduced the taxonomic ranks of class and species? | Caesalpino | John Ray | Tournefort | Linnaeus | ⒶⒷⒸⒹ |
| 4 | In 1665, Robert Hooke examined cork tissue under a microscope. What did he observe and describe as “cells”? | Chromosomes | DNA fibres | Empty chambers, thick walls | Mitochondria making ATP | ⒶⒷⒸⒹ |
| 5 | In a chloroplast, thylakoid stacks are called grana. The fluid part surrounding them is called: | Matrix | Stroma | Cristae | Lumen | ⒶⒷⒸⒹ |
| 6 | During the G2 phase of interphase, a cell mainly: | Replicates DNA | Forms spindle proteins, stores energy | Divides cytoplasm | Pairs homologous chromosomes | ⒶⒷⒸⒹ |
| 7 | Besides crossing over, which second event of meiosis produces genetic variation among gametes? | DNA replication | Independent combination of chromosomes | Spindle formation | Synapsis | ⒶⒷⒸⒹ |
| 8 | Which correctly matches the primary function of xylem and phloem in a plant? | Xylem—sugars; Phloem—water | Xylem—water; Phloem—sugars | Both—water only | Both—sugars & water | ⒶⒷⒸⒹ |
| 9 | Why do the respiratory and circulatory systems work closely together? | Both produce hormones | Circulatory delivers $O_2$, removes $CO_2$ | Both store minerals | Circulatory filters urine | ⒶⒷⒸⒹ |
| 10 | Which test confirms starch in a food sample, shown by a dark blue-black colour? | Biuret test | Benedict's test | Iodine test | Emulsion test | ⒶⒷⒸⒹ |
| 11 | In DNA, 30% of the bases are adenine. What percentage of the bases would be cytosine? | 20% | 30% | 40% | 70% | ⒶⒷⒸⒹ |
| 12 | Heating an enzyme far above its optimum temperature permanently damages its shape and active site. This is called: | Inhibition | Saturation | Denaturation | Activation | ⒶⒷⒸⒹ |

## SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt 11 questions. Choose either the main question OR its alternative.

| Part | Question | Marks | OR | OR Question | Marks |
|---|---|---|---|---|---|
| **2(i)** | State the four characteristics of a good scientific hypothesis, as mentioned in the textbook. | **3** | **OR** | What is Dengue fever? State its viral cause, its mosquito vector, and the recommended method of prevention, as described in the textbook. | **3** |
| **2(ii)** | Describe the contributions of any THREE early classifiers (before Linnaeus) mentioned in the textbook's history of classification: Andrea Caesalpino, John Ray, and Tournefort. | **3** | **OR** | Kingdom Protista includes very diverse organisms. Name the three types of protist-like organisms described in the textbook, with one example of each. | **3** |

| | | | | | |
|---|---|---|---|---|---|
| **2(iii)** | Describe the discovery of the cell by Robert Hooke in 1665, and explain how the discovery of the cell nucleus followed in 1831. | **3** | **OR** | Describe the structure of a chloroplast, including the roles of the chloroplast envelope, grana, and stroma. | **3** |
| **2(iv)** | State the chemical composition of the cell membrane according to the fluid mosaic model, including the role of cholesterol. | **3** | **OR** | Differentiate between chromoplasts and leucoplasts, including their location and function in a plant. | **3** |
| **2(v)** | Explain, according to the textbook, how crossing over and the independent combination of chromosomes during meiosis together produce genetic variation among gametes. | **3** | **OR** | A somatic (diploid) cell of an organism has 24 chromosomes. State how many chromosomes and how many chromatids per chromosome would be present at late prophase of mitosis (after S phase), and how many chromosomes each daughter cell would have once mitosis is complete. | **3** |
| **2(vi)** | Describe the role of the Golgi apparatus and the phragmoplast in forming the cell wall between daughter cells during cytokinesis of a plant cell. | **3** | **OR** | State the main events that take place during the G2 phase of interphase, according to the textbook. | **3** |
| **2(vii)** | Name the organs included in the digestive system, and state its role in maintaining homeostasis, as given in the textbook. | **3** | **OR** | What is meant by “organ systems working together”? Explain using the textbook's example of how the respiratory and circulatory systems cooperate. | **3** |
| **2(viii)** | State the six major bioelements of the human body along with their approximate percentages, as given in the textbook. | **3** | **OR** | Differentiate between purines and pyrimidines as nitrogen bases found in nucleic acids, naming the members of each group. | **3** |
| **2(ix)** | State any THREE benefits of cellulose fibre in the human diet and industry, as mentioned in the textbook. | **3** | **OR** | Describe the four levels of protein structure (primary, secondary, tertiary, and quaternary) as described in the textbook. | **3** |
| **2(x)** | Differentiate between the lock-and-key model and the induced-fit model of enzyme action, as referred to in the textbook's exercise. | **3** | **OR** | Explain why ATP is called the “energy currency” of the cell, and describe the ATP–ADP cycle. | **3** |
| **2(xi)** | Explain the term “energy of activation”. How do enzymes affect the activation energy required for a biochemical reaction? | **3** | **OR** | Explain how the rate of an enzyme-controlled reaction changes as substrate concentration increases, and what is meant by “saturation of active sites”. | **3** |

## SECTION C — Long Questions (4 × 5 = 20 Marks) — Attempt all 4

| Part | Question | Marks | OR | OR Question | Marks |
|---|---|---|---|---|---|
| **3(i)** | (a) What is meant by “science is a collaborative field”? Explain interdisciplinary research collaboration using any TWO examples from the textbook. [3]<br>(b) Name any THREE emerging careers in biology mentioned in the textbook that combine biology with other disciplines, briefly describing what each one studies. [2] | **5** | **OR** | (a) List any FOUR living characteristics of viruses AND any THREE non-living characteristics of viruses, as described in the textbook. [4]<br>(b) Why are viruses not included in any kingdom or domain of modern classification? [1] | **5** |
| **3(ii)** | (a) Describe the structure of mitochondria, including the role of cristae and the significance of mitochondria having their own DNA and ribosomes. [3]<br>(b) Describe the structure and function of cilia and flagella in eukaryotic cells. [2] | **5** | **OR** | (a) Describe, step by step, how a lysosome is formed and how it carries out intracellular digestion of a food vacuole, as shown in the textbook's diagram. [3]<br>(b) What is the role of the endoplasmic reticulum in transporting materials within the cell? [2] | **5** |
| **3(iii)** | (a) The diploid chromosome number of the fruit fly is 8. Using your understanding of meiosis, state how many chromosomes are present in each gamete, and explain why this reduction is necessary to maintain a constant chromosome number across generations. [3]<br>(b) Define a bivalent (tetrad) and state how many chromatids and centromeres it contains. [2] | **5** | **OR** | Explain why Meiosis II is often compared to mitosis. Identify any THREE specific similarities between the two processes. [5] | **5** |
| **3(iv)** | (a) Using a comparison table, give any FIVE differences between DNA and RNA. [3]<br>(b) If a double-stranded DNA molecule contains 20% guanine, calculate the percentage of adenine, thymine, and cytosine in the molecule, explaining your reasoning. [2] | **5** | **OR** | (a) Differentiate between intracellular and extracellular enzymes, giving ONE example of each mentioned in the textbook (other than pepsin). [2]<br>(b) Explain how the production and activity of enzymes within a cell can be regulated, according to the textbook. [3] | **5** |

Invigilator's Signature: ____________________ Candidate's Signature: ____________________

**Chapters Covered:** Ch 1 — The Science of Biology | Ch 2 — Biodiversity | Ch 3 — The Cell | Ch 4 — Cell Cycle | Ch 5 — Tissues, Organs & Organ Systems | Ch 6 — Molecular Biology | Ch 7 — Metabolism (Sec 7.1–7.6)

**— End of Question Paper —**